

Data Science & Machine Learning Project Catalog

11 Practical Capstones for Portfolios, Interviews, and Industry Readiness

Dayyan Hasan

Curriculum & Practical Projects Guide

Follow on LinkedIn: [dayyanhasan57](#)

Follow on GitHub: [@dynhsn](#)

Overview: This catalog outlines 11 curated, industry-relevant Machine Learning project ideas spanning supervised learning (classification and regression), unsupervised clustering, basic natural language processing (NLP), and recommendation systems. Each project is crafted to cultivate end-to-end data science competencies, including feature engineering, metric evaluation, and business interpretation.

1. Telco Customer Churn Prediction

Problem Focus: Predict whether a telecommunications subscriber will discontinue their service subscription.

Core Approach: Logistic Regression + Decision Tree.

Type & Difficulty: Classification | Beginner-friendly | Business & Retention focus.

Key Takeaway: Excellent starting project for understanding churn economics, probability thresholds, and baseline classification modeling.

2. House Price Prediction

Problem Focus: Estimate real estate transaction values using continuous housing attributes.

Core Approach: Linear Regression and regularized variants.

Type & Difficulty: Regression | Beginner-friendly | Real Estate Valuation.

Key Takeaway: Strong vehicle for mastering continuous evaluation metrics (RMSE, MAE, R^2) as a vital contrast to discrete classification metrics.

3. Employee Attrition Prediction

Problem Focus: Identify drivers behind employee turnover using organizational data (IBM HR Analytics dataset).

Core Approach: Supervised Classification (Logistic Regression, Random Forest).

Type & Difficulty: Classification | Intermediate | Human Resources Analytics.

Key Takeaway: Similar structural setup to churn prediction with a corporate HR perspective, ideal for candidates targeting enterprise and non-tech industries.

4. Loan Approval / Credit Risk Prediction

Problem Focus: Assess financial default probability and automate credit eligibility decisions from applicant profiles.

Core Approach: Tree-based models and logistic scoring models.

Type & Difficulty: Classification | Intermediate | FinTech & Risk Management.

Key Takeaway: Features messy real-world tabular data requiring rigorous missing-value imputation, categorical encoding, and feature engineering; a frequent FinTech interview staple.

5. Heart Disease / Diabetes Risk Prediction

Problem Focus: Detect the presence or likelihood of illness from clinical indicators (age, blood pressure, cholesterol, glucose level).

Core Approach: Probabilistic Classification and sensitivity analysis.

Type & Difficulty: Classification | Beginner-to-Intermediate | Healthcare & Clinical Diagnostics.

Key Takeaway: Demonstrates asymmetric classification risk where false negatives carry severe consequences, providing an essential practical link to Recall, Precision, and ROC-AUC optimization.

6. Mall Customer Segmentation

Problem Focus: Discover latent customer personas by grouping individuals based on annual income and spending score.

Core Approach: *K*-Means Clustering, Elbow Method, Silhouette Analysis.

Type & Difficulty: Unsupervised Learning | Beginner-friendly | Retail Marketing Analytics.

Key Takeaway: A strong non-classification capstone providing hands-on experience in unsupervised grouping, cluster profiling, and actionable business strategy formulation.

7. Sales Forecasting & Dashboard

Problem Focus: Analyze multi-region retail transaction histories across time and project forthcoming monthly revenue figures.

Core Approach: Exploratory Data Analysis, Trend/Seasonality Modeling, Linear/Ridge Regression, Visualization.

Type & Difficulty: Time-Series/Regression | Intermediate | Business Intelligence.

Key Takeaway: Centers on executive communication and storytelling, culminating in clear analytic dashboard summaries and forecasting insights.

8. Movie / Streaming Recommendation System (Content-Based)

Problem Focus: Suggest media titles aligned with viewer preferences by measuring genre, metadata, and rating similarity.

Core Approach: Cosine Similarity, TF-IDF on genre/tags, Feature Vectorization.

Type & Difficulty: Recommendation Systems | Beginner-to-Intermediate | Media & Entertainment.

Key Takeaway: A refreshing alternative to standard tabular prediction tasks, introducing similarity matrices without the overhead of full collaborative filtering.

9. Fake News / Spam Detection

Problem Focus: Differentiate between legitimate communications and misleading or unsolicited text (SMS / news headlines).

Core Approach: Text Vectorization (CountVectorizer, TfidfVectorizer) + Logistic Regression.

Type & Difficulty: Natural Language Processing (NLP) | Stretch Project | Content Moderation.

Key Takeaway: Broadens skillsets beyond structured tables into unstructured text processing, vocabulary modeling, and basic text cleaning.

10. Student Performance Prediction

Problem Focus: Model student exam outcomes and overall academic standing using study habits, attendance records, and socio-educational variables.

Core Approach: Regression or Binary Classification.

Type & Difficulty: Supervised Learning | Beginner-friendly | Education Analytics.

Key Takeaway: Highly intuitive domain context allowing students to speak with natural authority and clarity about feature importance and causality in job interviews.

11. Airline / Flight Delay Prediction

Problem Focus: Forecast flight delays by integrating scheduled flight plans, carrier data, route corridors, and meteorological observations.

Core Approach: Supervised Classification (e.g., Logistic Regression, Gradient Boosting).

Type & Difficulty: Classification | Intermediate-to-Advanced | Aviation Operations.

Key Takeaway: Exposes students to high-cardinality categorical variables, temporal dependencies, and noisy multi-source real-world datasets.

Quick Reference Matrix

#	Project Title	Learning Paradigm	Primary Evaluation Metric
1	Telco Churn	Classification	Accuracy, ROC-AUC, F1-Score
2	House Prices	Regression	RMSE, MAE, R^2
3	Employee Attrition	Classification	Recall, Precision, Confusion Matrix
4	Loan Approval / Credit Risk	Classification	ROC-AUC, PR-AUC, Gini
5	Heart Disease / Diabetes	Classification	Recall (Sensitivity), Specificity
6	Mall Customer Segmentation	Unsupervised	Silhouette Score, Inertia
7	Sales Forecasting	Regression / Analytics	MAPE, RMSE, Trend Charts
8	Streaming Recommender	Information Retrieval	Cosine Similarity, Top- K Accuracy
9	Fake News / Spam	NLP Classification	F1-Score, Precision
10	Student Performance	Classification / Regression	Accuracy, R^2
11	Flight Delay Prediction	Classification	Log-Loss, ROC-AUC